From Cash to Buy-Now-Pay-Later:
Platform Credit and Consumer Naivety

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Motivation

Buy-now-pay-later (BNPL, consumer credit) is increasingly used in e-commerce, and the platform hosting the purchase (Amazon, JD.com, etc.) supplies the credit itself.

- **The benefit.** Credit relaxes a **liquidity constraint**: a buyer without the cash on hand can trade.
- **The concern.** Credit carries a **hidden load** — late fees, penalty charges, credit-record damage — and the buyer sees only part of it at the checkout. She pays all of it.
- **Regulation is moving:** the UK brings BNPL into the FCA perimeter in July 2026; the EU's directive covers it from November 2026; Australia moved first.

This Paper

- **We ask** how **platform provision** and **consumer naivety** distort the provision of consumer credit — and what policy follows.
- **We model** BNPL's liquidity benefit and hidden load, with naive consumers, across three allocations: laissez-faire, a constrained planner, and a monopolist platform.

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- **We ask** how **platform provision** and **consumer naivety** distort the provision of consumer credit — and what policy follows.
- **We model** BNPL's liquidity benefit and hidden load, with naive consumers, across three allocations: laissez-faire, a constrained planner, and a monopolist platform.
- **Two takeaways:**
  - **Platform provision:** the platform's fees structure tend to titlt sellers toward credit — even the same net fee on cash and credit already over-supplies credit.
  - **Consumer naivety:** buyers see credit as cheaper than it is, so the market over-supplies it; naivety also makes them less price-elastic, amplifying the role of the platform's fee structure in credit over-provision.
- **Implication:** two policy tools are needed.

Roadmap

1. Environment and laissez-faire
2. The platform economy
3. Consumer naivety
4. Policy implications

Part 1 of 4

Environment and Laissez-Faire

Sellers and Buyers

- **Sellers.**
  - A unit mass indexed by quality $\xi \sim G$ on $[\xi_\ell, \xi_h]$; each is a monopolist with marginal cost $c\xi$.
  - Each seller meets a pool of buyers of measure one.
- **Buyers.**
  - A buyer is a pair (θ, y), her private information.
  - $\theta \sim F$ on $[0, \theta_h]$: willingness to pay per unit of quality.
  - $y \sim H$ on $[0, \infty)$: cash on hand for this purchase.
  - Unit demand: quality ξ at dollar price p gives utility

$$\theta\xi - p.$$

- **Interpreting quality.**
  - A higher ξ is a more valuable, more expensive good — performance, durability, brand, etc.

Payment and Demand

- **Payment.**
  - Money (M): any method that requires the full amount at the moment of purchase; every seller accepts money.
  - Credit (B): any pay-later method; it is seller's choice, accepting it costs seller $\omega \in (0, 1)$ of the transaction renenue.
  - If the seller accepts both, deferring is worth something, so she always pays by credit.

- **Demand.**
  - Let p be the dollar price, $z \equiv p/\xi$ the normalized price.
  - A money buyer must want the good ($\theta \geq z$) *and* be able to pay it ($y \geq p$); a credit buyer faces only the first hurdle:

$$d_M(z, \xi) = \overline{F}(z)\,\overline{H}(p); \qquad d_B(z) = \overline{F}(z),$$

    where $\overline{H}(p)$ is the fraction of buyers who can pay the dollar price.
  - At the same price, d_M is **both lower and more elastic**.

Seller Pricing in Laissez-Faire

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- Under **credit**, the seller solves

$$\max_z \left[(1-x)z - c\right] \overline{F}(z) = (1-x) \max_z (z-a)\overline{F}(z),$$

where x is an ad valorem fee ($x = \omega$ in laissez-faire) that turns unit cost c into effective cost $a \equiv c/(1-x)$. The Lerner condition gives

$$\frac{z_B - a}{z_B} = \frac{1}{\lambda(z_B)},$$

where $\lambda(z) \equiv zf(z)/\overline{F}(z)$ is the elasticity of credit demand.
- Profit per unit of quality, before the fee, is

$$\pi_B(a) = (z_B - a)\overline{F}(z_B).$$

The actual profit is $\xi\,\pi_B$.

Seller Pricing in Laissez-Faire

- Under **money**, demand is constrained by cash on hand:

$$(1-x)\max_z (z-a)\overline{F}(z)\overline{H}(z\xi),$$

and the Lerner condition is

$$\frac{z_M - a}{z_M} = \frac{1}{\lambda(z_M) + \gamma(p_M)}, \qquad p_M \equiv z_M \xi.$$

- $\gamma(p) \equiv p\, h(p)/\overline{H}(p)$ is the liquidity elasticity: a 1% rise in the dollar price leaves γ% of buyers unable to pay.

- Money profit per unit of quality, before the fee, is

$$\pi_M(a,\xi) = (z_M - a)\,\overline{F}(z_M)\,\overline{H}(p_M).$$

High-Quality Sellers Adopt Credit

- $\pi_B(a)$ is independent of ξ; $\pi_M(a,\xi)$ falls in ξ at exactly the liquidity elasticity:

$$\frac{\partial \ln \pi_M(c,\xi)}{\partial \ln \xi} = -\gamma(p_M) \ < \ 0.$$

Proposition

Suppose $\pi_M(c,\xi_h) < (1-\omega)\pi_B\left(\frac{c}{1-\omega}\right) < \pi_M(c,\xi_\ell)$. Then there is a unique ξ^* $\in (\xi_\ell, \xi_h)$ solving

$$\underbrace{(1-\omega)\,\pi_B\left(\frac{c}{1-\omega}\right)}_{\text{credit profit, flat in } \xi} = \underbrace{\pi_M(c,\xi^*)}_{\text{money profit, falling in } \xi},$$

sellers with $\xi > \xi^*$ accept credit; those with $\xi < \xi^*$ accept money only.

Compare Laissez-Faire to Planner

- The planner can assign sellers to payment rails, taking prices as given.
- With consumer surplus $S(z) \equiv \int_z^{\theta_h} (\theta - z)\, dF(\theta)$, total surplus is

$$W_M(\xi) = \pi_M + \overline{H}(z_M \xi) S(z_M), \quad W_B = (1-\omega)\pi_B + S(z_B),$$

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- At ξ^*, money and credit profits are equal, so

$$W_M(\xi^*) - W_B = \overline{H}(p_M^*) S(z_M^*) - S(z_B^*) :$$

  credit **expands the market** but **raises the price**.

Proposition

Laissez-faire over-supplies credit, $\xi^* < \xi^{**}$, iff

$$\overline{H}(p_M^*) S(z_M^*) > S(z_B^*).$$

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The Platform Economy

The Platform

- **Matching advantage.**
  - The platform brings each seller a pool of measure $m > 1$ buyers.
  - $m > 1$ supports positive platform fees through seller participation.
  - m multiplies profits on both rails equally, so it cancels from the money-vs.-pay-later choice.
- **Fees.**
  - The platform charges *ad valorem* fees: s_M on money, $\tilde{s}_B$ on credit; we work with the net credit fee

$$s_B = \frac{\tilde{s}_B - \omega}{1 - \omega} \approx \tilde{s}_B - \omega,$$

    so $s_B = s_M$ means equal net margins.
  - Strategy space: $(s_M, s_B) \in [0, \bar{s}_M(m)] \times [0, \bar{s}_B(m)]$.
  - Positivity suffices for most results, and holds whenever $m > 1$.

Sellers: Money or Credit?

- After paying the fee, a seller keeps a per-unit-quality profit of

$$V_B(s_B) = (1 - s_B)(1 - \omega)\, \pi_B\Big(\tfrac{c}{(1-s_B)(1-\omega)}\Big),$$
$$V_M(s_M, \xi) = (1 - s_M)\, \pi_M\Big(\tfrac{c}{1-s_M}, \xi\Big).$$

- V_M strictly decreases in ξ while V_B does not depend on ξ, so there is a thresohold $\hat{\xi}(s_M, s_B)$ defined by

$$V_B(s_B) = V_M(s_M, \hat{\xi}),$$

that the seller adopts credit iff $\xi \geq \hat{\xi}$.

When Does Platform Over-Supply Credit?

- Fix the reference seller: seller-ξ^*, who was exactly indifferent between credit and money under laissez-faire.
- Define the **log profit loss** after the platform's fees are imposed:

$$C_M \equiv \ln \frac{V_M(0,\xi^*)}{V_M(s_M,\xi^*)}, \qquad C_B \equiv \ln \frac{V_B(0)}{V_B(s_B)},$$

Then,

$$\hat{\xi} < \xi^* \iff C_M > C_B.$$

- The platform over-supplies credit (compared to laissze-faire) when seller-ξ^* suffers relatively more by moving from laissez-faire to platform at the per-unit-quality level.

What Goes into C_j?

- $C_j, j \in \{M, B\}$ is the cummulative profit loss caused by fee increase.
- By the envelope theorem,

$$\frac{\partial V_j(x)}{\partial x} = -R_j(x), \quad \frac{R_j}{V_j} = \frac{\varepsilon_j}{1-x},$$

  so

$$-\frac{d \ln V_j}{dx} = \frac{\varepsilon_j}{1-x},$$

  where R_j is sales revenue and ε_j is the demand elasticity.
- An elastic seller earns a thin margin, so the same fee destroys more of her profit.

Credit Oversupply: The Perspective of Elasticity

- **Log-distance.** With $t = \ln \frac{1}{1-s}$,

$$C_j = \int_0^{T_j} \varepsilon_j(t)\, dt = \bar{\varepsilon}_j\, T_j,$$

where $T_M = \ln \frac{1}{1-s_M}$ and $T_B = \ln \frac{1}{1-s_B}$

Theorem

At any fee pair (s_M, s_B) for which the threshold is interior,

$$\hat{\xi} < \xi^* \iff \bar{\varepsilon}_M\, T_M > \bar{\varepsilon}_B\, T_B \iff \frac{\bar{\varepsilon}_M}{\bar{\varepsilon}_B} > \frac{T_B}{T_M}.$$

Equal Net Fees Are Not Allocation-Neutral

- Cost-based regulation allows fee differences across rails only insofar as costs differ.

Proposition

Equal net fees $s_M = s_B = s$ give equal windows,

$$T_M = T_B = \ln \frac{1}{1-s} \equiv T,$$

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- With Pareto F, $\overline{F}(z) = \min\left\{1, (z/\theta_0)^{-\lambda}\right\}$:

$$\bar{\varepsilon}_M = \lambda + \bar{\gamma}_M \; > \; \lambda = \bar{\varepsilon}_B, \qquad \bar{\gamma}_M \equiv \frac{1}{T}\int_0^T \gamma\big(p_M(t)\big)\, dt,$$

and the threshold contracts to $\hat{\xi}(s) = (1-s)\,\xi^*$.

General Demand: Liquidity Against Drift

- For general F, the elasticity gap splits in two:

$$\bar{\varepsilon}_M - \bar{\varepsilon}_B \;=\; \underbrace{\bar{\gamma}_M}_{\text{liquidity}} \;-\; \underbrace{\overline{\Delta\lambda}}_{\text{drift}},$$

$$\overline{\Delta\lambda} \equiv \frac{1}{T}\int_0^T \left[\lambda(z_B(t)) - \lambda(z_M(t))\right] dt.$$

- **Drift.** Both rails read the same $\lambda(z)$ at *different* prices, and credit's is the higher one. Under Marshall's second law (λ nondecreasing), the drift works *against* over-supply.

- **Liquidity wins at small ω.** At $\omega = 0$,

$$\lambda(z_M) + \gamma(p_M) > \lambda\big(z^B(c)\big),$$

by continuity, the same holds for small $\omega > 0$, for all F, H, s.

General Fee Pair (s_M, s_B): Over-Supply Is a Region

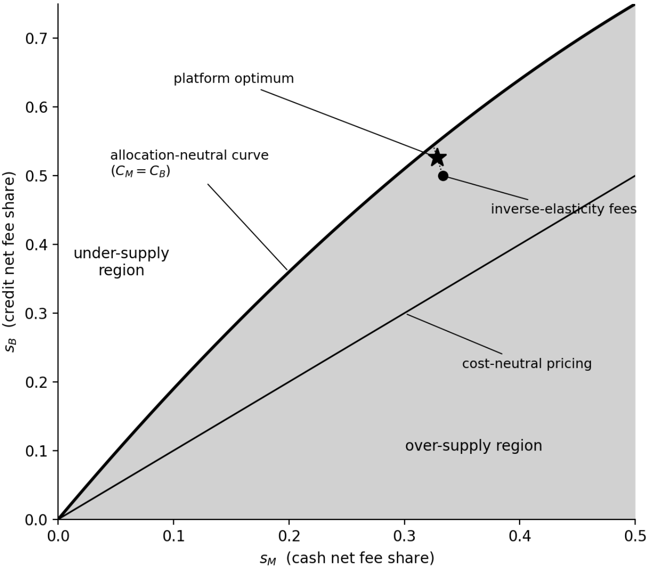
- Now let the platform post any pair, so $T_M \neq T_B$.

Proposition (the over-supply region)

Let $\overline{F}$ have constant elasticity. For any fee pair with interior threshold,

$$\hat{\xi} < \xi^* \iff s_B < 1 - (1 - s_M)^n, \qquad n \equiv 1 + \frac{\bar{\gamma}_M}{\varepsilon} > 1.$$

- **For small fees** the boundary is the ray $s_B \approx n\, s_M$: credit's net share must exceed money's by the elasticity ratio.



Parameters: $\varepsilon = 2,\ \bar{\gamma}_M = 2,\ n = 2.$

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Consumer Naivety

The True Credit Cost, and What the Buyer Perceives

- A credit buyer's true all-in price is

$$z^{\text{true}} = (1 - \beta + \mu)\, z,$$

where β is the deferral benefit per dollar — the free loan over the installment window — and $\mu > 0$ the expected hidden load per dollar — late fees, penalty charges, credit-record damage. We assume $\beta > \mu$: deferral is worth having.

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- A naive buyer perceives

$$\tilde{z} = \underbrace{(1 - \beta + v\mu)}_{\phi}\, z + \underbrace{(1 - v)k}_{\rho} = \phi z + \rho,$$

  where $v \in [0, 1]$ is the share of the load she perceives (the price-proportional part); the rest, weight $1 - v$, is her fixed belief k — a stored impression, lower than the true load.

Two Mistakes: Omission and Substitution

Naivety makes two connected mistakes: it omits part of the load and substitutes a fixed belief k.

- **Level.** The perceived price misses the true load by

  $$z^{\text{true}} - \tilde{z} = (1 - v)(\mu z - k) > 0.$$

  Buyers underestimate what credit costs.

- **Slope.** A one-percent rise in z moves $\tilde{z}$ by less than one percent,

  $$\frac{d \ln \tilde{z}}{d \ln z} = \frac{\phi z}{\phi z + \rho} < 1,$$

  Credit demand becomes less elastic.

- Naive buyers are not only blind to the load; they are deaf to its variation.

Naivety Leads to Credit Over-Provision in Laissez-Faire

- The laissez-faire threshold still solves

$$V_B(\omega; v) = V_M(0, \xi^*(v)),$$

As v falls, $V_B(\omega; v)$ rises — a less visible load lowers the perceived price and raises demand, while V_M does not involve v — so the threshold falls: $\xi^*(v) \downarrow$, more credit.

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- The market–planner comparison is the same race: expansion against price:

$$\xi^*(v) < \xi^{**}(v) \iff \overline{H}(p_M^*)\, S(z_M) \; > \; S_B^{\text{true}}(v).$$

Proposition

If μ is large enough, there is a cutoff $v^\dagger(\mu)$: for all $v < v^\dagger$, $\xi^* < \xi^{**}$.

Platform Over-Provision, Amplified by Naivety

- Cash demand elasticity is unchanged:

$$\varepsilon_M(z) = \lambda(z) + \gamma(z\xi).$$

- With naivety, credit demand becomes less elastic:

$$\varepsilon_B(z; v) = \lambda(\tilde{z}) \cdot \frac{\phi z}{\phi z + \rho} < \varepsilon.$$

- Under equal net fees $s_M = s_B = s$ (Pareto $\overline{F}$, arbitrary H), the platform over-provides credit at every $v \in [0, 1]$:

$$\frac{C_M}{C_B} = \frac{\bar{\varepsilon}_M}{\bar{\varepsilon}_B(v)} > 1.$$

 The wedge grows with naivety:

$$\frac{d}{dv}\left(\frac{C_M}{C_B}\right) \propto -\bar{\varepsilon}'_B(v) < 0.$$

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Policy Implications

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- **Education: raise v.**
  - Telling buyers that pay-later is costly is not enough.
  - What moves v is the **per-transaction, price-denominated disclosure**, e.g, "on this 600-dollar purchase, a missed installment costs you this much."
  - Generic warnings move at most k.
- **Both level and relative structure of platform fees matter.**
  - Only capping the fee level misses the real problem.
  - Platforms **subsidize** pay-later adoption — $s_B < s_M$, aggressively $s_B < 0$ — the most distortionary point in the fee plane.

The Regulatory Wave

So far, every major regime attacks the *consumer-side* problem — with education and affordability instruments — and none touches the *platform-side* problem: the fee structure of platform-provided credit.

- **United Kingdom.** The most v-friendly regime: FCA rules (in force July 2026) require per-purchase, amount-specific disclosure of repayment plans and missed-payment consequences, plus repayment reminders — all raise v; affordability checks and dispute/refund rights instead act on μ (who bears it, and how much of it realizes).
- **European Union.** The revised Consumer Credit Directive (applying November 2026) mandates standardized APR disclosure — a price-*proportional* cost statement, the form best matched to raising v — and creditworthiness assessments on μ's incidence.